

SoC Design Verification

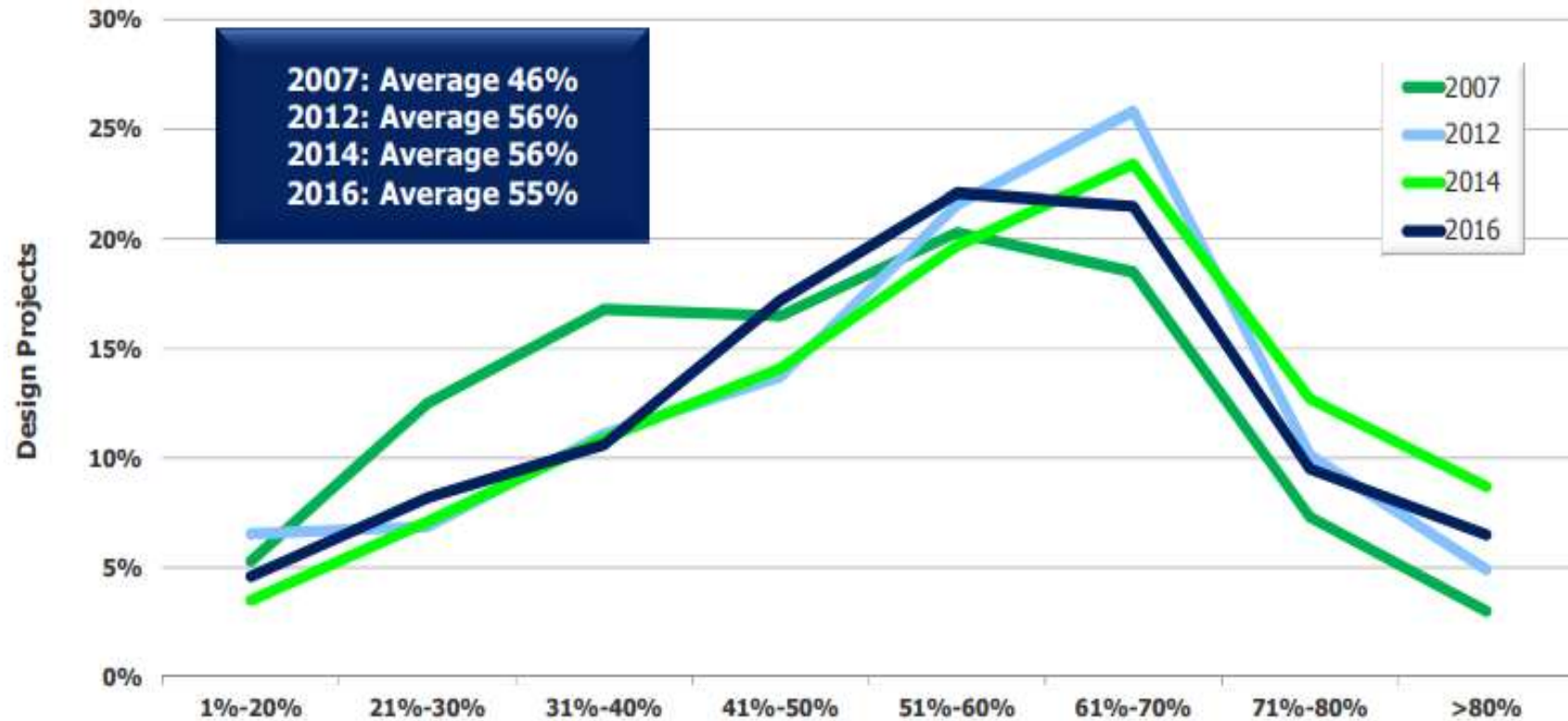
COE 838: Systems-on-Chip Design

What is SoC Design Verification?

- Design Verification is the process of checking that a given design correctly implements the specification.
- The largest task in SoC development that has the biggest impact on the key business drivers:
 - quality, schedule, design to market and cost.
- Verification team to Design team ratio ranges from 2:1 to 3:1
- > 70% development cycle is dedicated to verification.
- A diverse domain/field with endless strategies/techniques to sign-off (finalize) the chip.
- Every design needs verification - need for highly skilled design verification engineers to meet the challenges of the state-of-the-art technological innovations.

Verification Development Cycle Time

ASIC/IC Verification Project Time



SoC Design/Verification in Industry

- For Example: Advanced Micro Devices and Qualcomm are “Chip and SoC Makers”
- They make SoCs for living.
- Their SoCs are ranging in all computational applications.
- CPU, GPU, and all SoC elements that we can observe in:
 - Enterprise servers, Google, Facebook, Amazon, etc.
 - Smart Phones, Laptops, Desktops, Gaming Desktops.
 - Game consoles such as Xbox and Play-station.
 - Discrete Graphics Cards.

Verification Group in the SoC Developer - AMD

- RTG “Radeon Technologies Group” design and verify the AMD dGPU...
- dGPU : Discrete Graphics Processing Units
- Graphics in advanced applications of daily usage:
 - Virtual Reality, Augmented Reality, Gaming consoles/stations, Artificial Intelligence, Machine Learning..
- Compute graphics - a parallel processing computational engine convenient for certain applications than a typical sequential CPU....

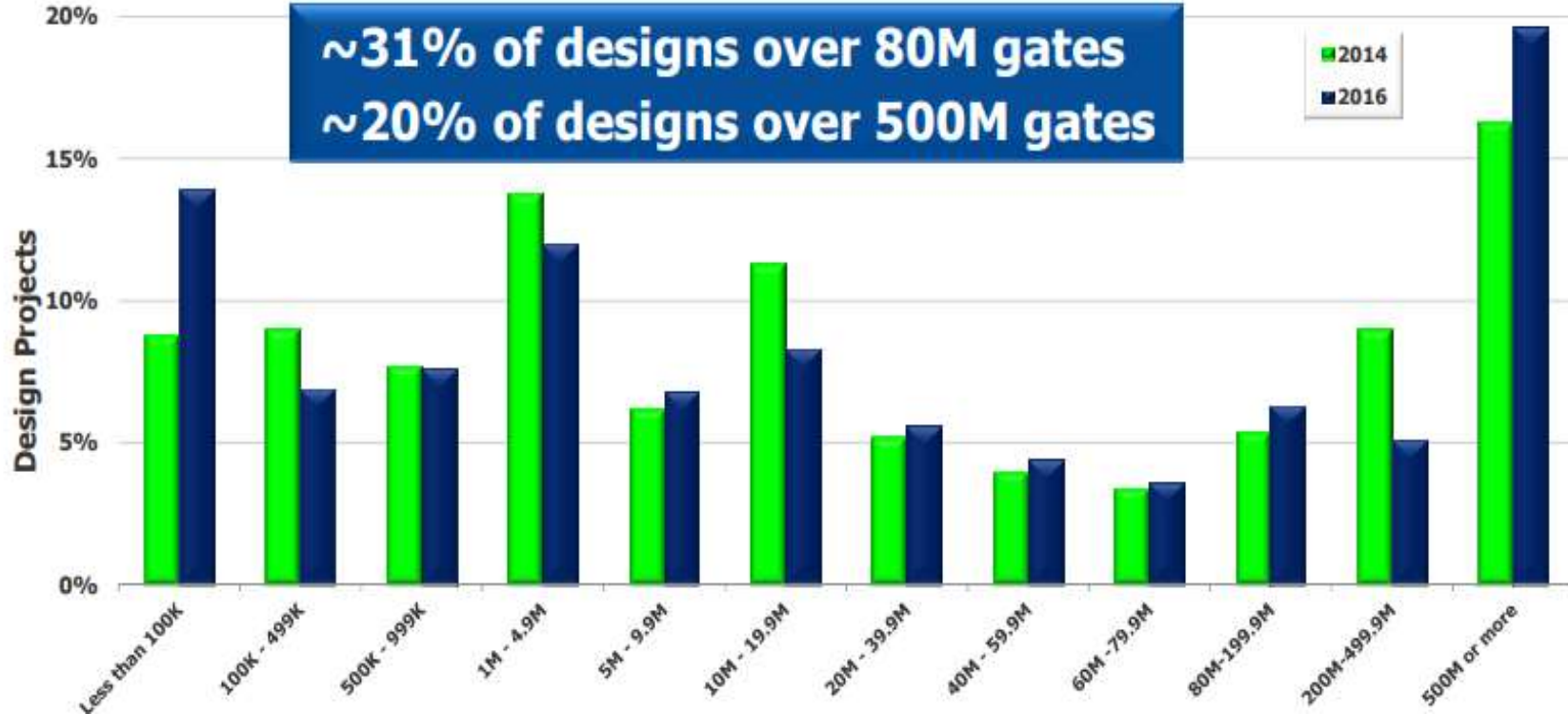
What is an SoC

- * A **system on a chip** or **system on chip** (**SoC** or **SOC**) is an [integrated circuit](#) (also known as an "IC" or "chip") that integrates all components of a [computer](#) or other [electronic](#) systems.
- * It may contain [digital](#), [analog](#), [mixed](#), and often [radio-frequency](#) functions all on a single [chip](#).
- * SoCs are very common in [mobile computing](#) market due to low power consumption.
- * Main applications are the area of [embedded systems](#).



SoC is too complex!

Design Sizes



Industry Always Look for Verification Engineers

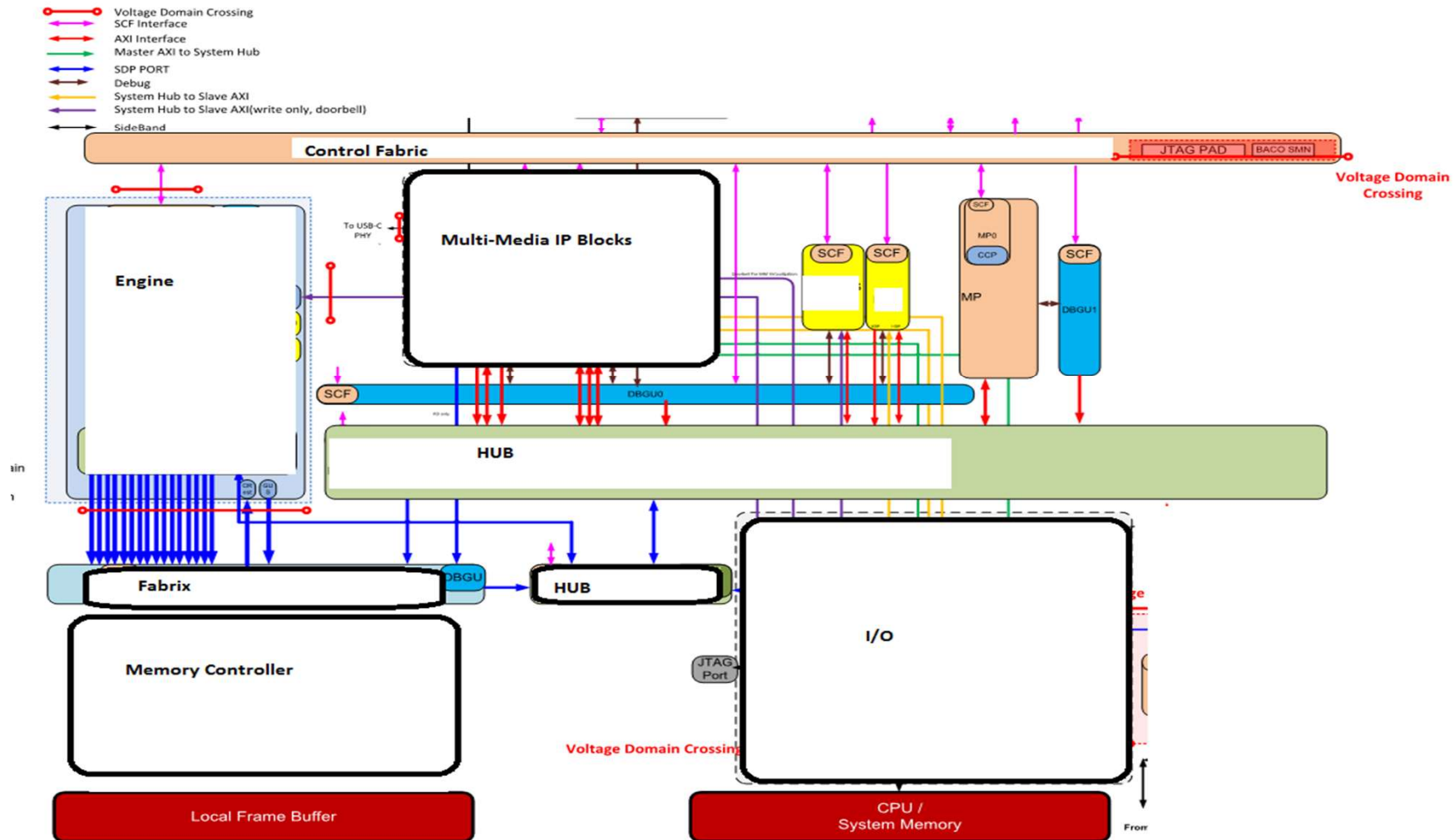


An SoC Architecture

Notes:

- Arrow out indicates master.
- Arrow in indicates slave.
- Bidir indicates master and slave capability.
- Lines passing under hubs are NOT connected to the hub.
- SCF stubs represent connectivity to SCF to simplify drawing

TOP VIEW

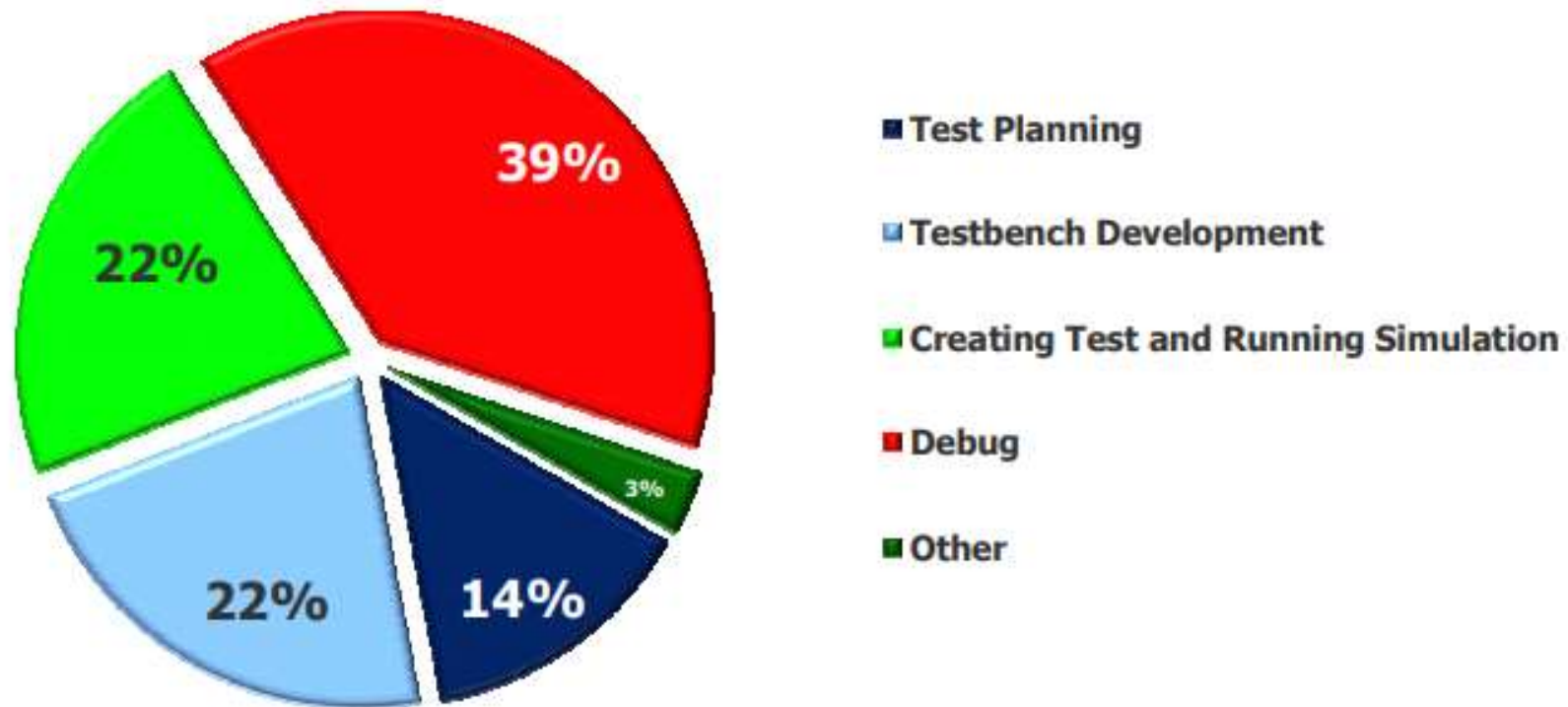


Verification Strategies

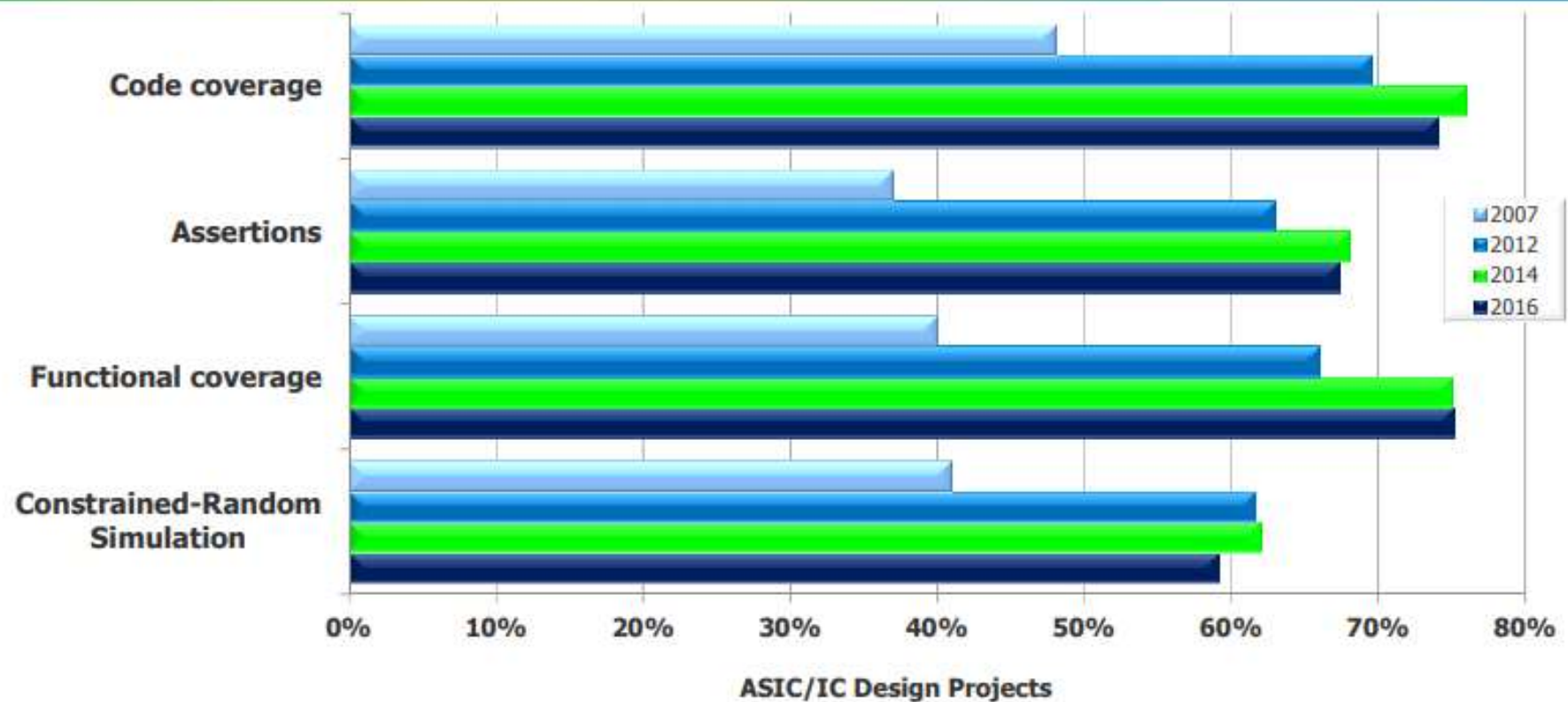
- **Functional Verification, Power Verification, Performance Verification**
- **Stimulus Generation**
- **Simulation based verification**
- **Directed testing**
- **Random and Constraint-Random Verification**
- **Formal Verification**
- **Verification Methodologies: UVM**
 - **FPGA-based Verification**
 - **Emulation**
 - **Verification tools and Infrastructure.**

Verification Engineers and their Time Consumption?

2016 Where ASIC/IC Verification Engineers Spend Their Time

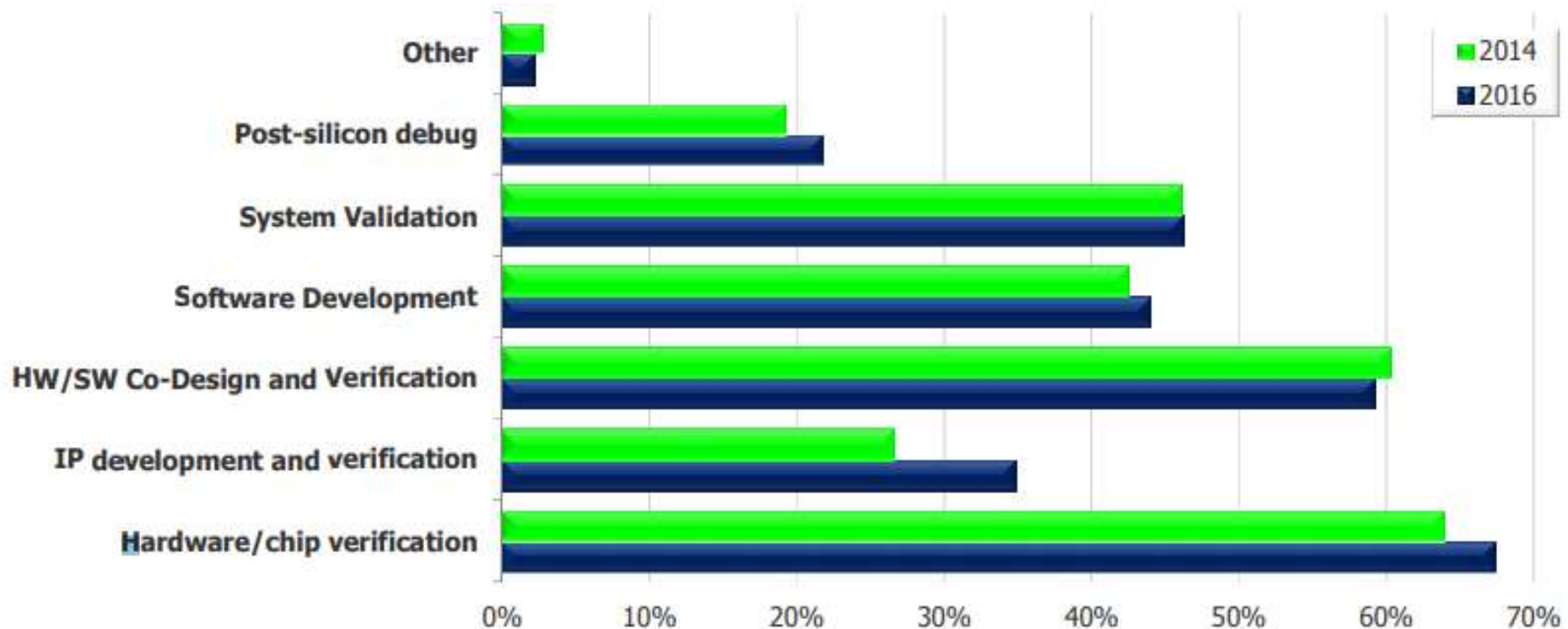


Mix & Match Verification Efforts and Trends!

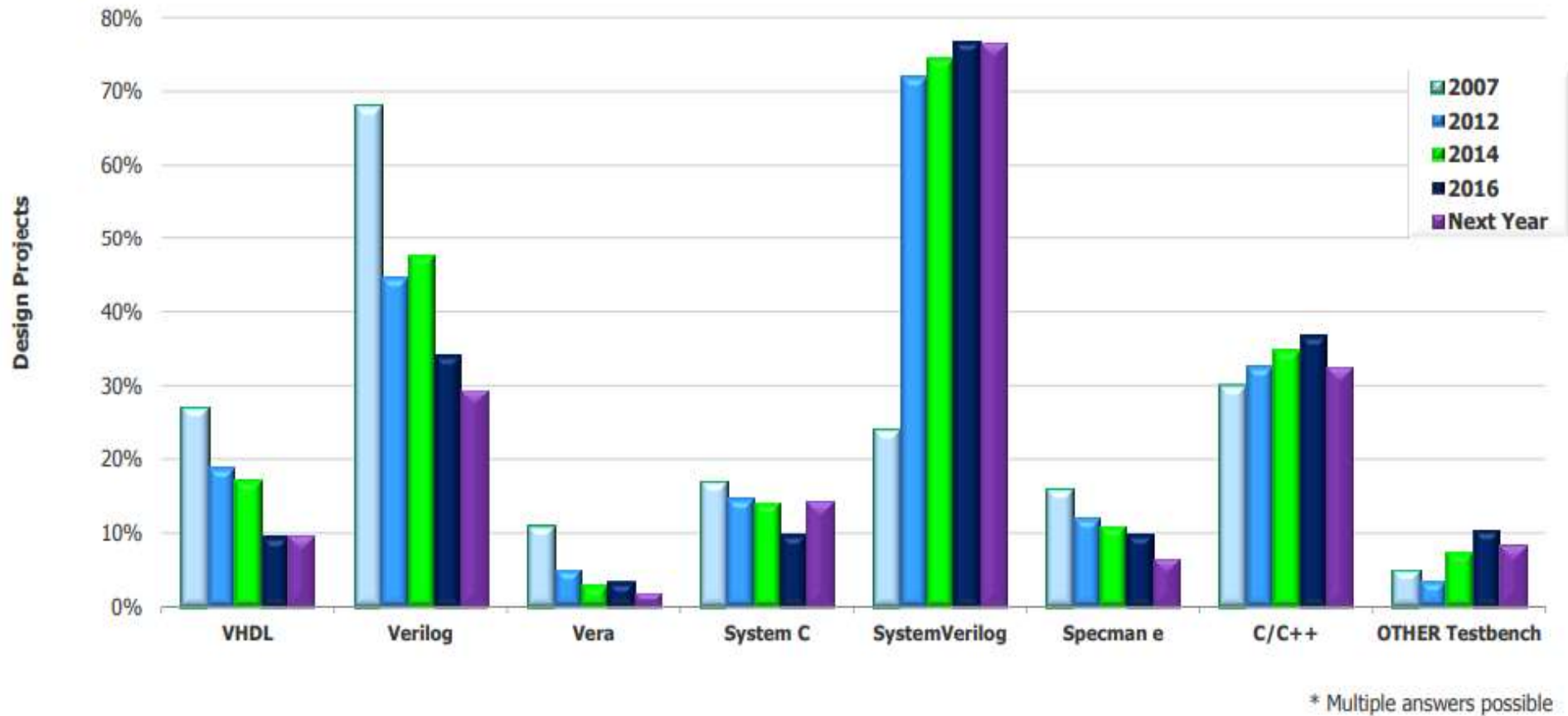


Acceleration is the Key

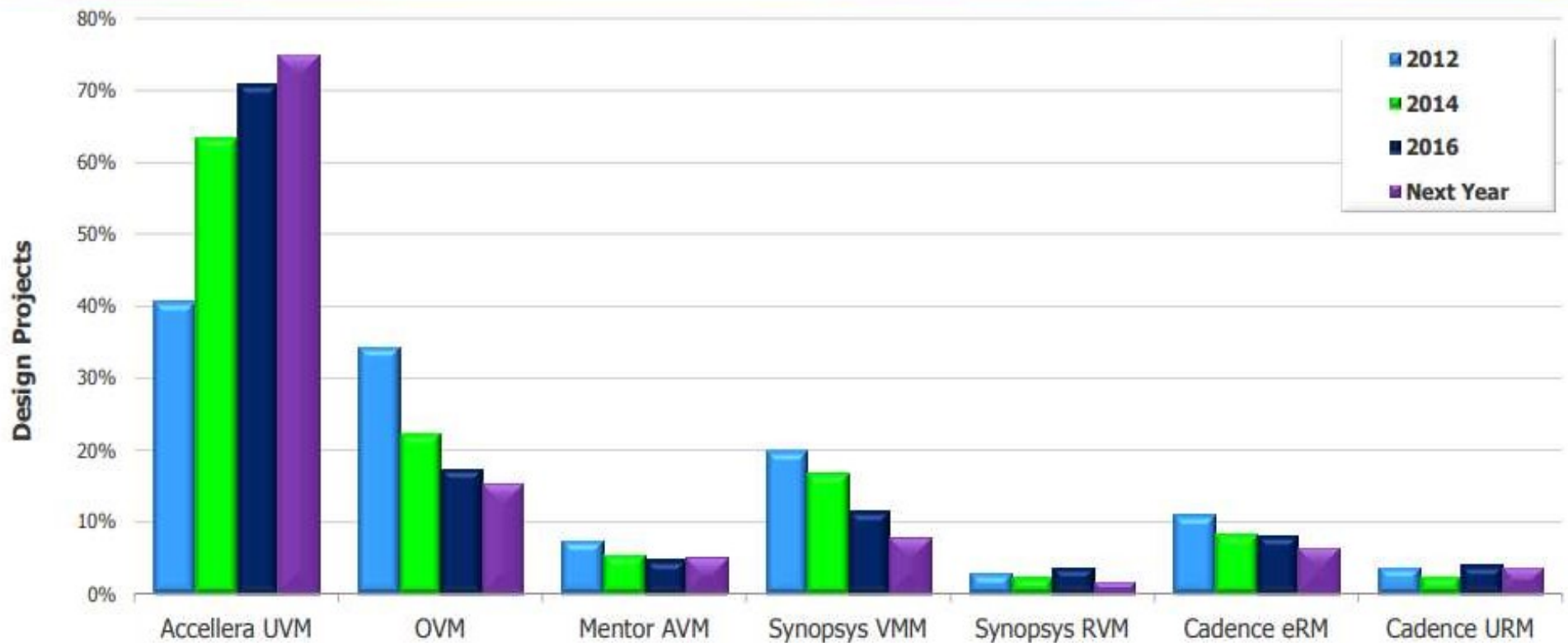
Emulation & FPGA Verification



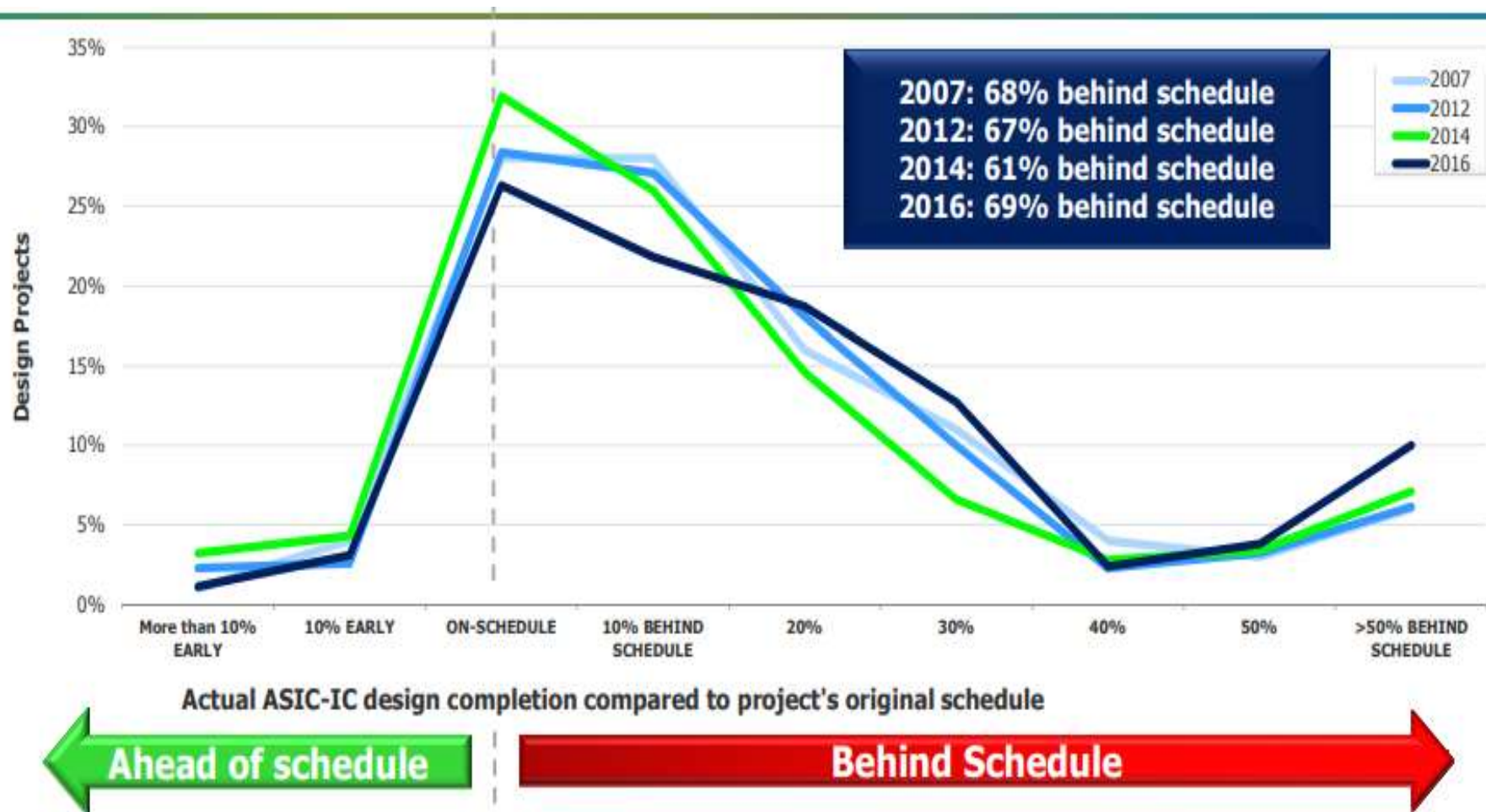
Verification Languages



Verification Methodologies Evolution....



SoC Design Projects Typical Schedules



How many times would it take to launch a product? Spins Headache...

